YOGABALAN B R

Embedded Systems • IoT • Full-Stack Development

Email: yogabalan2007yoga@gmail.com | GitHub: github.com/yogabalan07 | Location: Tamil Nadu, India

EDUCATION

KSR College of Engineering 2024 — 2028 | 3rd Year

Bachelor of Engineering — Computer Science and Engineering

Relevant Coursework: Data Structures, Microprocessors, Computer Networks, Operating Systems, DBMS, Embedded Systems

EXPERIENCE

Touchmark Descience Pvt Ltd — Web Development Intern Chennai, TN | 2026

- Collaborated in cross-functional agile team engineering responsive UI components with React & Tailwind CSS.
- Integrated secure REST API endpoints with Spring Boot microservices and PostgreSQL databases.
- Built critical modules for an Enterprise Inventory System, improving query performance and data accuracy.

TECHNICAL SKILLS

- Programming Languages: C, C++, Python, JavaScript (ES6+), TypeScript, SQL
- Embedded & Hardware: ESP32, Arduino, Raspberry Pi, LoRa SX1278 (433MHz), MPU6050 IMU, FreeRTOS, SPI/I2C/UART
- Web & Cloud: React, Vite, Node.js, Express, Tailwind CSS, PostgreSQL, Firebase, Supabase, REST APIs
- Developer Tools: Git, GitHub, VS Code, PlatformIO, Linux/Bash, Postman

FEATURED PROJECTS

ESP32 LoRa Morse Communicator (Hardware & RF)

Off-grid 433MHz transceiver with SX1278 modules transmitting encrypted text and Morse code audio over 3+ km without internet.

Tech: ESP32, LoRa SX1278, C++, FreeRTOS, SPI, I2C OLED

Two-Wheeled Self-Balancing Robot (Robotics & Control)

Inverted pendulum robot powered by discrete PID loop control and MPU6050 accelerometer/gyroscope complementary filtering.

Tech: Arduino Uno, MPU6050, PID Algorithm, PWM Motor Drivers, C++

CONNECT — Academic Discussion Platform (Full-Stack Web)

Campus forum for engineering students and faculty featuring Markdown/LaTeX math notes, verification workflows, and topic categorization.

Tech: React, TypeScript, Node.js, Express, PostgreSQL, Tailwind CSS

NRB Vidyalaya Learning Management System (EdTech)

Hindi literacy and pronunciation learning platform with interactive phonetic audio synthesis, quizzes, and teacher grading dashboards.

Tech: React, Vite, Node.js, Firebase Firestore, Web Audio API

Enterprise Business Management System

Inventory, purchase ledger, and vendor analytics system with ACID compliant schemas and automated invoice PDF generation.

Tech: React, TypeScript, Spring Boot, PostgreSQL, Tailwind CSS

HONORS & ACHIEVEMENTS

- Ship Hackathon — First Prize (Intra-College Event)
- Build 2 Pitch — Second Prize, Cash Reward Received (Intra-College Event)